

Industrial Safety Intelligence Platform

Overview

Industrial Safety Intelligence Platform is an AI-powered solution designed to prevent industrial accidents by identifying compound risks before they escalate into critical incidents.

The system combines data from multiple industrial sources such as IoT sensors, work permits, worker tracking systems, SCADA records, and regulatory standards to generate proactive safety recommendations.

Problem Statement

Traditional industrial safety systems operate independently and fail to correlate multiple risk factors occurring simultaneously. As a result, dangerous situations may remain undetected until an accident occurs.

Proposed Solution

Our platform introduces a Compound Risk Detection Engine that:

- Correlates multiple safety signals
- Detects compound hazards
- Predicts lead time before incidents
- Generates risk scores
- Provides intervention recommendations
- Supports regulatory compliance

Key Features

Compound Risk Detection

Identifies hazardous combinations of:

- Gas accumulation
- Active work permits
- Worker presence
- Confined space operations
- Maintenance activities

Lead Time Prediction

Provides early warning signals before risk reaches critical levels.

Risk Heatmap

Visualizes high-risk industrial zones in real time.

Intervention Engine

Generates actionable recommendations such as:

- Evacuation alerts
- Entry lockdowns
- Permit suspension
- Safety escalation procedures

Compliance Intelligence

References OISD, DGMS, and Factory Act guidelines.

Technology Stack

- Python
- Streamlit
- Pandas
- NumPy
- Plotly

System Workflow

1. Multi-source data ingestion
2. Compound risk evaluation
3. Trend analysis
4. Lead time prediction
5. Risk scoring
6. Alert generation
7. Intervention recommendation

Project Structure

|—— app.py |—— requirements.txt |—— README.md |—— architecture.png |——
documentation.pdf |—— presentation.pptx

Run Locally

```
pip install -r requirements.txt
```

```
streamlit run app.py
```

Expected Impact

- Reduced industrial accidents
- Faster incident response
- Improved worker safety
- Better regulatory compliance
- Lower false-negative risk detection

Future Scope

- Real-time IoT integration
- CCTV-based computer vision analytics
- Multi-agent AI orchestration
- Predictive maintenance modules

Team

ET AI Hackathon 2026 Submission

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